Nishchay Patel

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EDUCATION

Master of Engineering, Industrial and Systems Engineering 2023
Virginia Tech, Blacksburg, VA

Bachelor of Engineering, Mechanical Engineering 2021
L.J. Institute of Engineering and Technology (degree awarded by Gujarat Technological University), Ahmedabad, India
CGPA 8.65/10, first class with distinction

SELECTED GRADUATE COURSEWORK

- Statistics and analytics: Mathematical Probability and Statistics for Industrial and Systems Engineers (ISE 5034), Statistical Quality Control (ISE 5474), Data Analytics (CS 5525)
- Operations research: Optimization (ISE 5405), Decision Analysis for Engineers (ISE 5834), Production Systems Analysis (ISE 5044)

RESEARCH INTERESTS

- AI-driven versus conventional processes in operations research: where learned models improve planning and inventory decisions, and where classical methods still win
- Pretrained time-series models in operations research: when a model pretrained on other series, reading a short history as context with no per-series fitting, supports a stocking decision as well as a fitted model, and how to tell a zero-shot gain that generalizes from one memorized in pretraining
- Decision-aligned evaluation of intermittent and lumpy demand forecasts: scoring a forecast by simulating the inventory policy it feeds, with attention to out-of-sample design, model selection, target leakage, simulator initialization, and calibration under service-level constraints
- Demand estimation from censored records: when sales or issues are limited by the stock on hand, the observation process depends on the inventory policy
- Reproducible benchmarking of classical, machine-learning and pretrained methods on public data

PREPRINTS AND MANUSCRIPTS

All sole-authored.

1. **Evaluation design decides the winner: horizon, leakage and metric effects in intermittent demand forecasting** (2026).
   Preprint. https://doi.org/10.5281/zenodo.23004062. Version 2, October 7, 2026. Version 1, posted August 3, 2026, was titled "The conditional value of artificial intelligence in supply-chain demand planning: a reproducible out-of-sample benchmark across the intermittency spectrum." Public data (M5, UCI Online Retail II). Full pipeline and per-series results released; see ai-demand-planning-benchmark below.
   Benchmarks eight classical forecasters, a global gradient-boosted model (LightGBM), two deep global models (NHITS and DeepAR) and two pretrained zero-shot models (Chronos-Bolt and TimesFM) on the M5 competition data (30,490 store-SKU series) and UCI Online Retail II (4,675 e-commerce series) under one rolling-origin protocol. Changing only the evaluation design reorders the methods. Scoring cumulative lead-time demand instead of one step ahead lifts Croston's method on M5 from last of eight classical methods to fifth at a 13-week lead time. The study's initial Online Retail II specification used same-week transacted-price features, which leak the target and carried 42 percent of the gradient-boosted model's total gain. Restricting them to past values drops the model from ahead of all eight classical methods to behind five of them. Replacing MASE with RMSSE turns the naive benchmark from a frequent winner on Online Retail II into the method with the worst mean RMSSE on both

panels. On Online Retail II the method favored by in-sample fit loses to a naive forecast out of sample. Neither pretrained zero-shot model beats the best trained model in any sparse demand class on either panel.

2. **Evaluation pitfalls in intermittent-demand forecasting and inventory policy: a fleet-scale study of how far accuracy, calibration, and cost disagree** (2026).
   Preprint. https://doi.org/10.5281/zenodo.23222105. Anonymized operational data, available on request. An earlier version of the study is public with anonymized data. The simulator is specified in the paper, and code for this version will be released on journal publication.
   Benchmarks eleven forecasters, from the operator's three deployed methods to gradient boosting and a pretrained zero-shot model, on all 191,613 items with at least two nonzero training quarters in a large multi-brand maritime operator's spare-parts catalog, scored on point error and on simulated cost under an (s, S) inventory policy. The per-item Kendall tau-b between point-error rank and simulated-cost rank is 0.35 on demand-active items and 0.18 to 0.25 on items with demand in at least two of the four test quarters, so accuracy predicts cost only moderately. Safety stock set from the predictive quantiles of zero-inflated Poisson, hurdle negative binomial and gradient-boosting models did not beat a normal-approximation formula at matched fleet fill rate, and the best-calibrated model lost the most. On the four ships for which the operator's deployed (s, S) parameters are available, the deployed policy was cheaper than a simple exponential smoothing challenger at a higher fill rate under a symmetric start with a burn-in, reversing the ranking an asymmetric start had produced. Scoring on the most recent quarters, where recorded demand is far sparser and rewards under-forecasting, roughly halves the simulated cost of the classical methods tested and changes which one is cheapest.
3. **Navigating altmetrics: insights into data consistency, provenance, and applications** (2026).
   Under review.

RESEARCH SOFTWARE

ai-demand-planning-benchmark 2026

`https://github.com/NISHCHAY026/ai-demand-planning-benchmark`

Reproducible benchmark behind the evaluation-design paper. It holds the code that regenerates every table and figure of version 2 of the preprint from raw public data, the per-series results, and a consistency script that checks the manuscript's table values and key inline claims against the result files in more than 470 checks. Version 2 corrects data-construction defects found after version 1 was posted, and the repository documents each correction.

PROFESSIONAL EXPERIENCE

Technical Demand Planning Data Analyst September 2023 to Present

Holland America Group, Seattle, WA (remote)

- Tested the in-house best-fit model selection on 12 held-out quarters. The pipeline's own in-sample scores ranked SBA best of its three candidate methods (SBA, exponential smoothing and a short moving average) on 78 percent of series, but out of sample SBA was best on 40 percent and had the highest mean absolute scaled error of the three, worse than a naive forecast.
- Designed an order rationalization model that flagged excess purchase commitments. Buyers reviewed the flagged commitments and canceled about $24 million of them across fiscal years 2024 and 2025.
- Built SKU-level velocity segmentation that sets minimum, maximum and reorder quantities across roughly 1.8 million SKUs.
- Refactored more than 80 stored procedures behind the SQL and Azure forecasting pipelines so that runs are reproducible and auditable, and moved more than 50 BI assets off TIBCO and SQL Server, rebuilding their underlying data models on Snowflake.
- Led data integration for an enterprise planning system rollout and turned the planning logic into validated production datasets.

- Designed and prototyped a Power Apps application for logging and tracking corrective action plans on inventory-control, service, safety, maintenance and audit findings, on a nine-table Excel data model hosted on SharePoint, with a reviewer dashboard and a companion tracking workbook.
- Supervised interns in SQL development, data modeling and supply-chain analytics through to delivered dashboards and datasets.

Data Analyst August 2019 to June 2021
Radixweb, Ahmedabad, India

- Built and tuned ETL pipelines feeding a data warehouse for a global financial services client.
- Ran preprocessing and analysis on large transactional datasets in SQL and Python.

TEACHING

Instructional Assistant, Math Emporium January 2022 to June 2023
Virginia Tech, Blacksburg, VA

- Tutored more than 150 students per semester in linear algebra, calculus, business calculus and precalculus, and ran small-group sessions.
- Trained incoming tutors.

OTHER OPEN-SOURCE PROJECTS

- premarket-trader (github.com/NISHCHAY026/premarket-trader). Premarket momentum day-trading model for US equities, with walk-forward, out-of-fold machine-learning trade selection feeding an event-driven backtester and documented limits (few out-of-sample trades, daily bars only, survivorship bias).

TECHNICAL SKILLS

Programming and tooling. Python, pandas, NumPy, matplotlib, SQL, AMPL, C#, Git, LaTeX

Machine learning and pretrained models. LightGBM, scikit-learn, PyTorch, NeuralForecast (NHITS, DeepAR), Chronos-Bolt, TimesFM

Forecasting and inventory methods. Croston, Syntetos-Boylan Approximation, TSB, exponential smoothing, temporal aggregation (ADIDA, MAPA), rolling-origin evaluation, MASE and RMSSE scoring, (s, S) and base-stock inventory simulation

Statistical methods. Kendall tau-b rank correlation, bootstrap intervals (item, cluster and paired), rank-based multiple comparisons (Nemenyi, MCB) with clustered critical distances, zero-inflated Poisson and hurdle negative binomial count models, pinball loss, coverage calibration of predictive quantiles

Data and analytics platforms. Snowflake, SQL Server, Azure, Parquet, Power BI, Power Apps, Tableau, Kinaxis Maestro